

Exploratory Data Analysis (EDA) Report

Project Title: Wine Quality Analysis

Introduction

In this project, I performed **Exploratory Data Analysis (EDA)** on the **Red Wine Quality** dataset. The main aim of this project is to understand the dataset, clean the data, analyze different features, and find the relationship between alcohol content and wine quality. EDA helps us understand the data before building a machine learning model.

Objective

The objectives of this project are:

- To understand the Wine Quality dataset.
- To check the dataset structure and data types.
- To identify missing values and duplicate records.
- To calculate statistical measures.
- To detect and remove outliers.
- To visualize the data using different graphs.

Dataset Description

First, I imported the required libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Scikit-learn**. Then I loaded the **Red Wine Quality** dataset using Pandas. I used functions like `head()`, `tail()`, `shape`, `columns`, `info()`, and `describe()` to understand the dataset. These functions helped me know the number of rows, columns, data types, and summary statistics.

Data Cleaning

After loading the dataset, I checked for missing values using `isnull().sum()`. I also checked for duplicate records using `duplicated()`. Then I calculated the correlation between the numerical columns. Outliers in the **Alcohol** column were detected using the **Interquartile Range (IQR)** method and removed from the dataset to improve data quality.

Statistical Analysis

I calculated the following statistical measures for the numerical columns:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation Matrix

These statistical values helped me understand the distribution and relationship of different features in the dataset.

Data Analysis

I analyzed the relationship between different variables and wine quality. I sorted the correlation values to identify which features have the highest impact on wine quality. I also filtered wines with a quality score greater than **7** and sorted the dataset based on alcohol content to understand the quality of high-alcohol wines.

Data Visualization

To understand the dataset better, I created the following graphs:

- **Bar Chart** – Shows the number of wines in each quality category.
- **Histogram** – Displays the distribution of alcohol content.
- **Scatter Plot** – Shows the relationship between alcohol content and wine quality.
- **Line Chart** – Displays the variation in wine quality across the dataset.

These visualizations helped me understand the patterns and trends present in the dataset.

Observations

From the analysis, I observed the following:

- The dataset was loaded successfully.
- No missing values were found in the dataset.
- Duplicate records were checked.

- Outliers in the alcohol column were identified and removed.
- Alcohol content showed a positive relationship with wine quality.
- Most wines belonged to the medium quality categories.
- The visualizations helped in understanding the distribution of alcohol and quality ratings.

Conclusion

This project helped me understand the importance of Exploratory Data Analysis in a machine learning project. I learned how to inspect the dataset, clean the data, calculate statistical measures, detect outliers, analyze correlations, and create different visualizations. After completing the EDA process, the dataset became clean and ready for building machine learning models to predict wine quality. This project also improved my understanding of data preprocessing and visualization techniques.